

Ex:No: 7	Implementation of Linear Regression Using mtcars Dataset
Date:	

Problem Description:

The mtcars dataset contains specifications of different car models, including fuel efficiency and engine characteristics. To understand how engine and vehicle attributes influence fuel efficiency, linear regression can be used to model the relationship between **miles per gallon (mpg)** and predictors such as **engine displacement (disp)**, **horsepower (hp)**, and **weight (wt)**.

Objective:

To implement a linear regression model in R to analyze the relationship between fuel efficiency and vehicle attributes, interpret the regression coefficients and model summary, and visualize the regression results for better understanding of the model.

```

1
2 library(caret)
3 library(ggplot2)
4 library(scatterplot3d)
5
6 # Step 2: Load Dataset
7 data(mtcars)
8
9 head(mtcars)
10 str(mtcars)
11
12 # Step 3: Split Dataset (70% Train, 30% Test)
13 set.seed(123)
14
15 trainIndex <- createDataPartition(mtcars$mpg, p = 0.7, list = FALSE)
16
17 train_data <- mtcars[trainIndex, ]
18 test_data <- mtcars[-trainIndex, ]
19
20 # Step 4: Simple Linear Regression (mpg ~ hp)
21 model_simple <- lm(mpg ~ hp, data = train_data)
22
23 summary(model_simple)
24
25 # Step 5: Predictions (Simple Model)
26 predictions_simple <- predict(model_simple)
27
28 head(predictions_simple)
29
30 # Step 6: Model Evaluation
31 mse <- mean((test_data$mpg - predictions_simple)^2)
32 print(paste("MSE:", mse))
33
34 r_squared <- summary(model_simple)$r.squared
35 print(paste("R-squared:", r_squared))
36
37 # Step 7: Visualization (hp vs mpg)

```

```

22
23 summary(model_simple)
24
25 # Step 5: Predictions (Simple Model)
26 predictions_simple <- predict(model_simple)
27
28 head(predictions_simple)
29
30 # Step 6: Model Evaluation
31 mse <- mean((test_data$mpg - predictions_simple)^2)
32 print(paste("MSE:", mse))
33
34 r_squared <- summary(model_simple)$r.squared
35 print(paste("R-squared:", r_squared))
36
37 # Step 7: Visualization (hp vs mpg)
38 ggplot(train_data, aes(x = hp, y = mpg)) +
39   geom_point(color = "blue", size = 3) +
40   geom_smooth(method = "lm", se = FALSE) +
41   ggtitle("Linear Regression: HP vs MPG")
42
43 # Step 8: Multiple Linear Regression (mpg ~ wt + hp + disp)
44 model_multiple <- lm(mpg ~ wt + hp + disp, data = train_data)
45
46 summary(model_multiple)
47
48 # Step 9: Prediction (as you requested - on training data)
49 predicted_multiple <- predict(model_multiple)
50 head(predicted_multiple)
51 # Step 10: Visualization (Multiple Variables)
52 scatterplot3d(mtcars$wt, mtcars$hp, mtcars$mpg,
53   main = "3D Scatter Plot: MPG vs Weight and HP",
54   xlab = "Weight",
55   ylab = "Horsepower",
56   zlab = "Miles per Gallon"
57 )
58 |

```

```

> head(mtcars)
      mpg cyl  disp  hp  drat   wt  qsec vs  am  gear  carb
Mazda RX4     21.0   6  160  110  3.90 2.620 16.46 0   1    4    4
Mazda RX4 Wag  21.0   6  160  110  3.90 2.875 17.02 0   1    4    4
Datsun 710     22.8   4  108   93  3.85 2.320 18.61 1   1    4    1
Hornet 4 Drive  21.4   6  258  110  3.08 3.215 19.44 1   0    3    1
Hornet Sportabout 18.7   8  360  175  3.15 3.440 17.02 0   0    3    2
Valiant        18.1   6  225  105  2.76 3.460 20.22 1   0    3    1

```

```

> summary(model_simple)

Call:
lm(formula = mpg ~ hp, data = train_data)

Residuals:
    Min       1Q   Median       3Q      Max
-5.0566 -2.0595 -0.8412  1.2261  8.8266

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 30.09557    1.82793   16.464 7.43e-14 ***
hp          -0.07141    0.01191   -5.996 4.92e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.9 on 22 degrees of freedom
Multiple R-squared:  0.6204,    Adjusted R-squared:  0.6031
F-statistic: 35.95 on 1 and 22 DF,  p-value: 4.919e-06

```

```
> print(paste("MSE:", mse))
[1] "MSE: 44.7806034917258"
> r_squared <- summary(model_simple)$r.squared
> print(paste("R-squared:", r_squared))
[1] "R-squared: 0.620374955565965"
> head(predictions_simple)
      Mazda RX4      Datsun 710      Hornet 4 Drive Hornet Sportabout
      22.24053      23.45449      22.24053      17.59892
      Valiant      Merc 240D
      22.59758      25.66819
```

```
> summary(model_multiple)
```

Call:

```
lm(formula = mpg ~ wt + hp + disp, data = train_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-4.1039	-1.6123	-0.0093	1.3078	5.5539

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	38.974631	2.593486	15.028	2.33e-12 ***
wt	-4.741613	1.342959	-3.531	0.0021 **
hp	-0.030224	0.012852	-2.352	0.0290 *
disp	0.003784	0.011662	0.324	0.7490

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.616 on 20 degrees of freedom

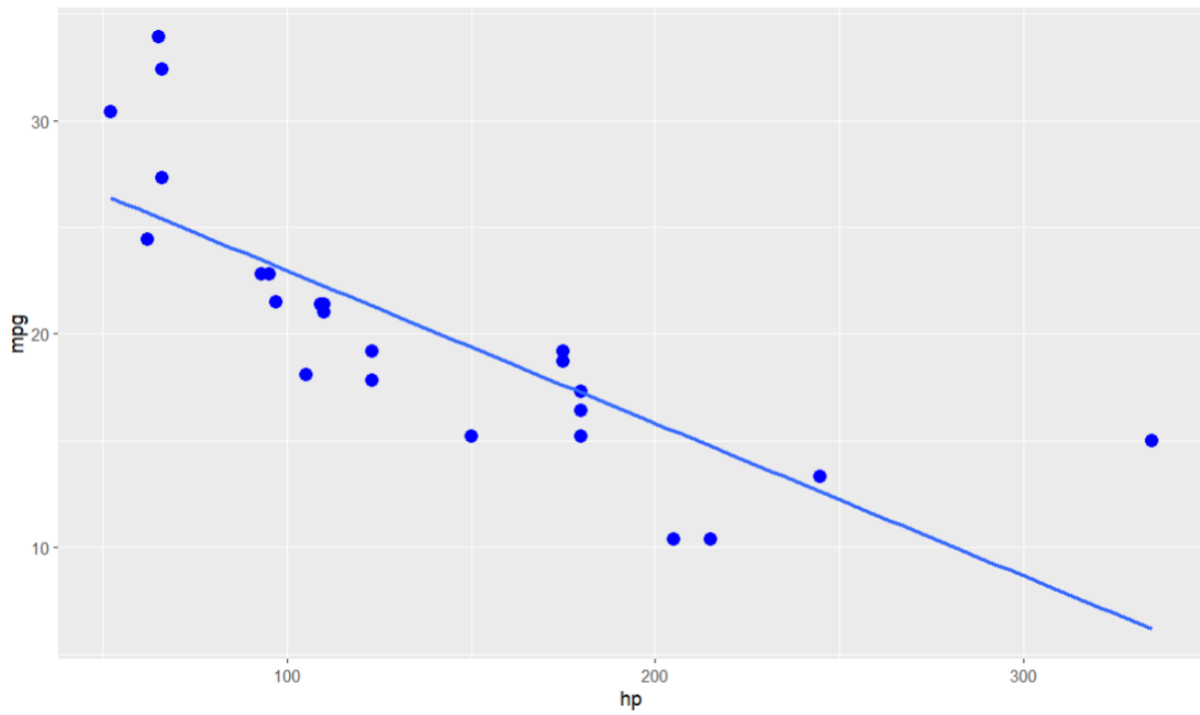
Multiple R-squared: 0.8447, Adjusted R-squared: 0.8214

F-statistic: 36.27 on 3 and 20 DF, p-value: 2.794e-08

```
> head(predicted_multiple)
```

Mazda RX4	Datsun 710	Hornet 4 Drive	Hornet Sportabout
23.83241	25.57193	21.38198	18.73652
Valiant	Merc 240D		
20.24653	22.53011		

Linear Regression: HP vs MPG



3D Scatter Plot: MPG vs Weight and HP

